

# Los Angeles Temperature-Forecasting Laboratory

*A Reproducible Audit of Deep Sequence Models, Weather Regimes,*

*Uncertainty Calibration, and Operational Forecast Governance*

Salem Morelli

Independent PhD Research Laboratory

Statistics

August 24, 2026

**Repository of record:** salemmorelli1/Los-Angeles-Daily-Temperature-High; source-code audit snapshot: main at commit 59135f1.

## Abstract

This report audits a production-oriented laboratory for daily maximum-temperature forecasts at Los Angeles International Airport (KLAX; 33.9425° N, 118.4081° W). The system combines Open-Meteo historical and forecast fields, National Weather Service (NWS) telemetry, a three-state hidden Markov weather-regime model, engineered atmospheric features, an LSTM or Transformer forecaster, an XGBoost sleeve, Monte Carlo dropout, split-conformal calibration, rule-based governance, and realized-error attribution. The committed artifact snapshot contains 3,157 uninterrupted daily records from January 1, 2018 through August 23, 2026; 129 predictors; chronological train/validation/test sizes of 2,206/473/473; and explicit one-, three-, and five-day target dates.

The principal statistical finding is a separation between holdout model skill and operational publication skill. On the chronological test set, LSTM mean absolute errors (MAEs) were 3.14°F, 4.06°F, and 4.48°F, while XGBoost achieved 2.30°F, 3.76°F, and 4.38°F. In the 65-row live attribution tape, the pre-anchor model achieved 1.68°F, 3.09°F, and 3.37°F MAE; however, canonical forecasts after NWS anchoring recorded 7.18°F, 7.56°F, and 7.04°F MAE. The older benchmark field exhibited an 8°F cold bias, so frequent anchoring transmitted benchmark error into the published forecast. The NWS grid-point self-check is now corrected and passing, but historical comparisons remain source-contaminated.

Conformalized test coverage was 92.6%, 87.6%, and 84.8% against a nominal 90%, whereas raw MC-dropout coverage deteriorated with horizon. The system is technically sophisticated and reproducible, yet the audit supports disabling benchmark anchoring until a clean post-correction evaluation establishes incremental value. Keywords: temperature forecasting, LSTM, XGBoost, hidden Markov model, conformal prediction, forecast governance.

## Research Objective and Audit Questions

The laboratory is best understood as a longitudinal prediction system rather than a single fitted model. At each feature date  $t$ , the estimand is the maximum temperature  $Y$  at valid date  $t+h$ , with  $h$  in  $\{1, 3, 5\}$ . The implementation must therefore preserve three clocks: observation time, feature time, and target-valid time. The report evaluates whether data, learning, uncertainty, and publication layers preserve that separation.

$$Y_{-}(t+h) = \max \text{ temperature on calendar date } (t + h), \quad h \in \{1, 3, 5\} \quad (1)$$

Five questions organize the audit. First, do all targets and realized outcomes share one calendar-day contract? Second, are features available at issuance time, including lagged regime probabilities, so that no future information leaks backward? Third, do temporal holdout errors demonstrate horizon-dependent generalization? Fourth, do predictive intervals attain useful coverage under explicitly stated calibration semantics? Fifth, does operational anchoring improve realized forecast accuracy relative to both the unanchored model and simple baselines?

**Inference boundary:** The repository artifacts support reproducible descriptive and predictive evaluation of this implemented pipeline. They do not establish causal effects of meteorological variables, universal superiority over official forecasts, or station-wide generalization beyond the KLAX-centered data contract.

All numerical findings below were recomputed or transcribed from versioned artifacts on the main branch. No synthetic quantity is reported as operational performance. Figures use the committed metrics directly; architecture diagrams are explanatory. The distinction matters because an attractive browser laboratory can communicate assumptions but does not replace the Python pipeline, cached data, or chronological evaluation.

## Data Sources, Location, and Measurement Contract

Part 0 constructs the canonical daily table from Open-Meteo and NWS services. Open-Meteo provides coordinate-indexed historical and forecast variables, while the NWS API provides official point forecasts and KLAX observations through JSON-LD services (National Weather Service, 2026; Open-Meteo, 2026a, 2026b). The artifact snapshot reports 3,157 daily rows, 22 normalized historical variables, 500 recent station observations, seven NWS forecast days, and seven Open-Meteo forecast days. Calendar continuity is explicitly true with zero gap days.

| Source / layer      | Role                     | Audited contract                     |
|---------------------|--------------------------|--------------------------------------|
| Open-Meteo archive  | Historical predictors    | 2018-01-01 to 2026-08-23; 3,157 days |
| Open-Meteo forecast | Exogenous outlook        | Seven daily forecast values          |
| NWS LOX grid        | Official benchmark       | LOX/148,41; live point check passed  |
| NWS KLAX            | Recent observations      | 500 station observations             |
| Part 0 parquet      | Canonical research table | 22 fields; continuous calendar       |

The coordinate contract is scientifically consequential. A grid cell several points inland can experience a different marine-layer and heat regime than the airport. The repository now queries the NWS points endpoint and fails loudly if the hardcoded office/grid tuple drifts. That correction converts a hidden spatial-validity assumption into an executable data test.

## Temporal Alignment and Leakage Control

The corrected target clock is  $\text{target\_date\_h} = \text{feature\_date} + h$  calendar days. Part 1 retains the unlabeled live tail, whereas model fitting drops rows missing any future target. Part 2 builds the final live sequence from the newest feature row; Part 9 backfills realized values against stored target-date columns. This contract prevents the common error of treating a forecast issuance date as though it were the feature date.

$$D_{\text{train}} = \{(X_t, Y_{(t+1)}, Y_{(t+3)}, Y_{(t+5)}) : \text{all three } Y \text{ values observed}\} \quad (2)$$

$$X_{\text{live}} = X_{(\text{max observed feature date})}, \quad \text{labels may be unavailable} \quad (3)$$

| Split      | Date interval            | Rows           | Use                        |
|------------|--------------------------|----------------|----------------------------|
| Training   | 2018-01-01 to 2024-01-15 | 2,206          | Fit parameters and scalars |
| Validation | 2024-01-16 to 2025-05-02 | 473            | Tune and calibrate         |
| Test       | 2025-05-03 to 2026-08-18 | 473            | Chronological holdout      |
| Live tail  | through 2026-08-23       | 5<br>unlabeled | Issue current forecast     |

The split is ordered, not randomized. This is necessary because serial correlation makes random partitions optimistic: neighboring days share seasonal, synoptic, and lagged information. The remaining concern is model-selection reuse of the validation interval. Conformal diagnostics partly reuse validation residuals, so the repository correctly treats the full-validation bands as display/risk bands rather than strict publishable intervals.

## End-to-End Architecture



Target clock:  $\text{target\_date\_h} = \text{feature\_date} + h \text{ calendar days} \mid H = 1, 3, 5$

Figure 1. Audited repository execution order and target clock.

The orchestrator runs Part 0, Part 6, Part 1, Part 2A, Part 2, optional Parts 2B and 2C, Part 3, and Part 9. This order is not cosmetic. Regimes must exist before lagged regime features; the alpha module must update feature metadata before model inference; governance must inspect the canonical forecast and interval state; attribution must wait until target dates mature.

Operationally, the architecture has two distinct products. The pre-anchor model is a statistical estimate based on learned predictors. The canonical product is a policy-mediated forecast that may blend the LSTM with XGBoost and then move that blend toward NWS. Both are retained in the prediction log. That paired logging makes the central audit possible: post-processing can be evaluated as an intervention on the forecast rather than being conflated with model accuracy.

## Feature Engineering and Atmospheric Alpha

The final model matrix contains 129 predictors. Feature families include contemporaneous daily summaries, temperature lags at 1, 2, 3, 5, 7, and 14 days, rolling means/variances/ranges over 3 to 30 days, cyclic calendar terms, pressure and humidity lags, wind-direction sine/cosine transforms, diurnal range, dew-point depression, precipitation, and regime probabilities. Part 2A adds pressure tendency and acceleration, marine-layer flags and streaks, cloud anomalies, humidity trends, wind acceleration, temperature momentum, 30-day z scores, time from recent peak, and ENSO proxy fields.

$$x_{\text{wind},\sin} = \sin(2\pi d/360), \quad x_{\text{wind},\cos} = \cos(2\pi d/360) \quad (4)$$

$$z_{\text{t},30} = (T_{\text{t}} - \text{mean}(T_{\text{t-29:t}})) / \text{sd}(T_{\text{t-29:t}}) \quad (5)$$

Circular wind encoding avoids the false Euclidean distance between 359° and 1°. Rolling and lagged variables summarize persistence and short-lived fronts, while the HMM posterior provides a lower-dimensional synoptic context. Regime fields are shifted one day before merging. The feature builder also removes constants and redundant physical probability columns when they duplicate statistical regime probabilities.

**Audit implication:** With 129 predictors and only 2,206 training rows, flexible learners face a nontrivial p/n ratio. Regularization, early stopping, stable scaling, and honest temporal holdouts are therefore core statistical requirements, not implementation preferences.

## Hidden Markov Weather-Regime Engine

Part 6 fits a three-state Gaussian hidden Markov model to 13 daily meteorological features. Let  $S_t$  denote the latent regime and  $X_t$  the observed weather vector. The likelihood factors through an initial state distribution, transition matrix  $A$ , and state-conditional Gaussian emissions.

$$p(x_{1:T}, s_{1:T}) = \pi(s_1) \prod_{t=2..T} A(s_{t-1}, s_t) \prod_{t=1..T} p(x_t | s_t) \quad (6)$$

The fitted model converged with log likelihood -31,331.52. Physical names are assigned only when rule scores cross a validation threshold: Regime 0 is validated as marine layer, Regime 2 as dry-clear, and Regime 1 remains unvalidated. This restraint is statistically appropriate; post hoc labels are interpretive hypotheses, not identified latent truths (Rabiner, 1989).

| State | Share | High °F | Humidity | Cloud | Physical label           |
|-------|-------|---------|----------|-------|--------------------------|
| 0     | 19.4% | 63.1    | 79.8%    | 63.3% | Marine layer (validated) |
| 1     | 44.7% | 72.9    | 79.2%    | 34.5% | Unvalidated              |
| 2     | 36.0% | 68.1    | 64.1%    | 33.1% | Dry-clear (validated)    |



## Regime Persistence and Mixing

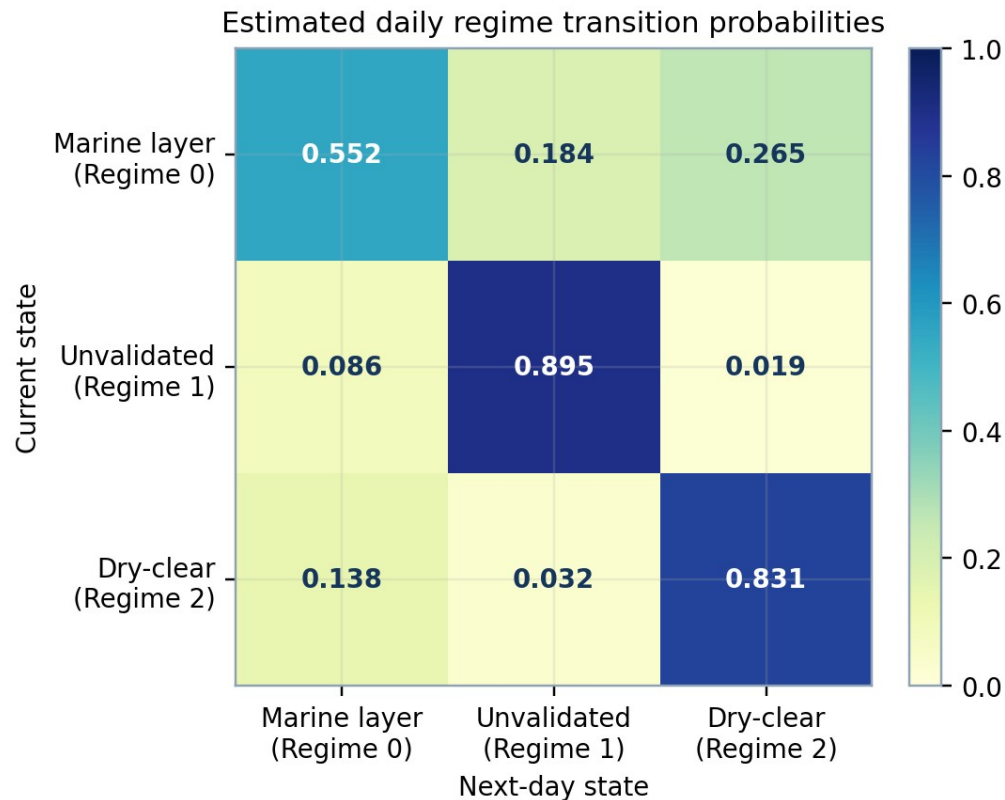


Figure 2. Estimated transition matrix for the three daily weather regimes.

Regime 1 and Regime 2 are strongly persistent, with self-transition probabilities .895 and .831.

Regime 0 is less persistent (.552) and has material one-day transitions into Regime 2 (.265). These estimates justify regime-aware forecasting because the state contains information about both today and tomorrow, but they also imply that a single global error distribution may understate conditional risk.

For inference, posterior state probabilities are preferable to hard labels. A hard Viterbi state treats classification as certain and creates discontinuities; posterior probabilities propagate ambiguity into downstream models. The repository supplies both one-hot states and selected probabilities, lagged by one day. Future work should estimate conditional MAE and interval coverage by regime, then require minimum effective sample sizes before any regime-specific publication rule is activated.

## Deep Sequence Forecast Formulation

The default learner is a two-layer LSTM with 128 hidden units, 20% dropout, a 14-day sequence, and three output heads for  $H = 1, 3$ , and  $5$ . LSTM gates regulate persistence through the cell state, mitigating the vanishing-gradient behavior of ordinary recurrent networks (Hochreiter & Schmidhuber, 1997). PyTorch autograd optimizes a horizon-weighted squared-error objective.

$$i_t = \sigma(W_i x_t + U_i h_{(t-1)} + b_i), \quad f_t = \sigma(W_f x_t + U_f h_{(t-1)} + b_f) \quad (7)$$

$$c_t = f_t \odot c_{(t-1)} + i_t \odot \tanh(W_c x_t + U_c h_{(t-1)} + b_c) \quad (8)$$

$$L(\theta) = \sum_h w_h \cdot \text{mean}[(Y_{(t+h)} - \hat{Y}_{(t+h)}(\theta))^2], \quad w = (1, .8, .6) \quad (9)$$

A Transformer alternative uses encoder blocks and self-attention, but the committed model is LSTM. The distinction matters because the report evaluates artifacts, not an unrealized architecture. Both model classes expose compatible three-horizon outputs. The sequence length provides two weeks of context; longer seasonal memory enters through engineered rolling and calendar features rather than recurrent state alone.

**Heat branch:** An asymmetric warm/heat loss remains as an experiment but is disabled. The committed audit found worse global MAE and zero of five test heat events predicted at or above 85°F, so standard horizon-weighted loss remains the production objective.

## Training Dynamics and Regularization

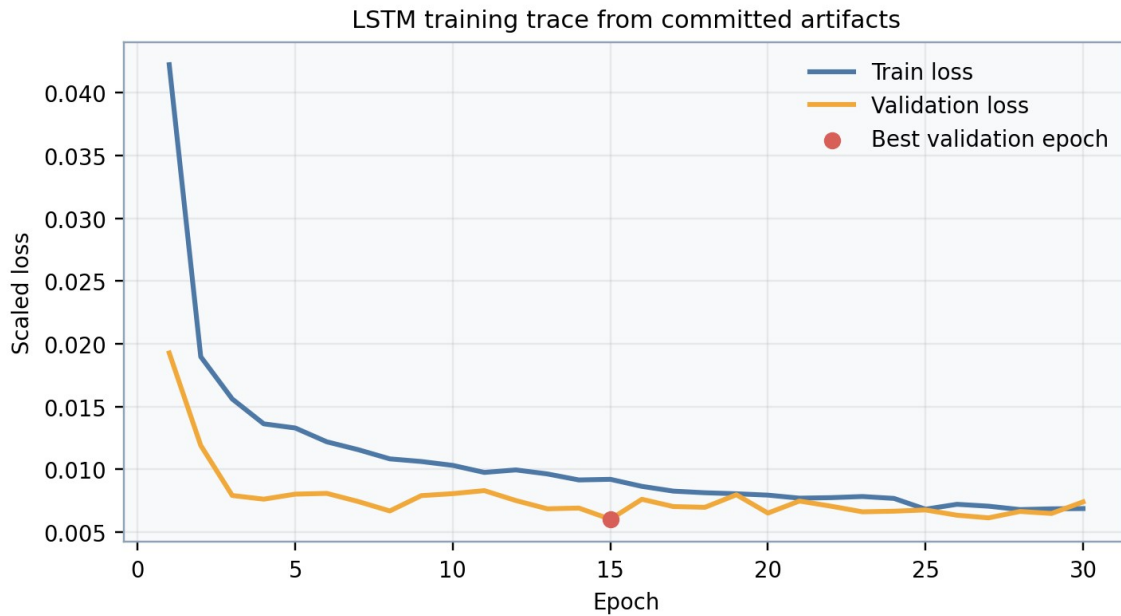


Figure 3. LSTM training and validation traces from the committed `training_history.json` artifact.

Training ran 30 epochs with early-stopping patience 15 and a learning-rate reduction from .001 to .0005 after epoch 23. The best validation loss (.00603, scaled units) occurred at epoch 15. Later training loss continued to decline while validation loss oscillated, a familiar signature of increasing estimation variance rather than continued generalization.

The stored feature and target scalers are fitted on the training period and reused for validation, testing, and live inference. This is essential: re-estimating scaling parameters on the full series would leak future distribution information. Reproducibility is strengthened by a fixed seed (42), committed model and scaler artifacts, metadata describing the 129-feature schema, and validators that reject feature-schema drift before predict-only operation.

## Chronological Holdout Results

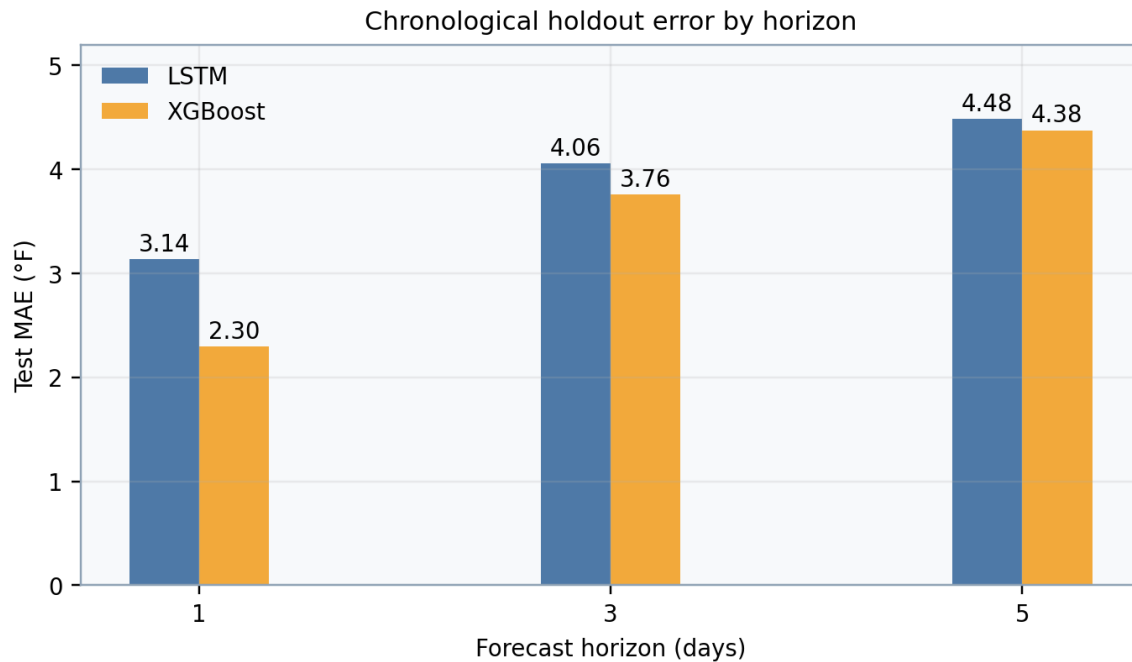


Figure 4. Test-set MAE by horizon for LSTM and XGBoost.

| Model   | H=1 MAE | H=3 MAE | H=5 MAE | Average |
|---------|---------|---------|---------|---------|
| LSTM    | 3.136   | 4.057   | 4.484   | 3.892   |
| XGBoost | 2.297   | 3.761   | 4.376   | 3.478   |

Errors increase with horizon, as expected when the future atmospheric state becomes less

constrained. XGBoost improves on the LSTM at every horizon, most at H=1. The comparison is not a complete model-selection test because the XGBoost sleeve uses the same chronological partitions and may have been tuned after observing LSTM performance. Nevertheless, the paired artifact suggests that tabular lag/rolling predictors capture substantial short-horizon signal without requiring recurrent state.

## Ensembling and the NWS Anchor

Part 2B blends LSTM and XGBoost and can then anchor the blend toward NWS. For horizon  $h$ , the pre-anchor blend is a convex combination, after which a policy-dependent anchor weight  $a_h$  may be applied.

$$B_h = w_h \text{XGB}_h + (1-w_h)\text{LSTM}_h \quad (10)$$

$$F_h = (1-a_h)B_h + a_h \text{NWS}_h, \quad 0 \leq a_h \leq 1 \quad (11)$$

The August 24 artifact used XGBoost weights .95, .40, and .40. NWS anchoring was not applied at  $H=1$ , but soft anchoring shifted  $H=3$  by 3.65°F and  $H=5$  by 3.85°F. The final forecast was 82.97°F, 83.00°F, and 80.43°F for August 24, 26, and 28. A heat guard additionally raised  $H=3$  because NWS predicted 85°F.

| H | LSTM  | XGB   | Pre-anchor | NWS  | Final | Anchor $\Delta$ |
|---|-------|-------|------------|------|-------|-----------------|
| 1 | 81.51 | 83.05 | 82.97      | 82.0 | 82.97 | 0.00            |
| 3 | 78.24 | 81.01 | 79.35      | 85.0 | 83.00 | +3.65           |
| 5 | 74.99 | 78.97 | 76.58      | 83.0 | 80.43 | +3.85           |

Anchoring is a decision rule, not a statistically free improvement. If the benchmark has location or retrieval bias, equation 11 transmits that bias in direct proportion to  $a_h$ .

## Uncertainty Estimation and Conformal Calibration

Part 2C performs 200 stochastic forward passes with dropout active, producing a Monte Carlo approximation to model uncertainty (Gal & Ghahramani, 2016). Raw dropout intervals were under-dispersed, especially at  $H=5$ . The code therefore applies split-conformal residual quantiles. If  $q_{(1-\alpha)}$  is the finite-sample quantile of absolute calibration residuals, a symmetric interval is centered on the canonical forecast.

$$q_{(1-\alpha)} = \text{Quantile}_{\{\text{ceil}((n+1)(1-\alpha))/n\}} \{|Y_i - \hat{Y}_i|\} \quad (12)$$

$$C_h(x) = [F_h(x) - q_h, F_h(x) + q_h] \quad (13)$$

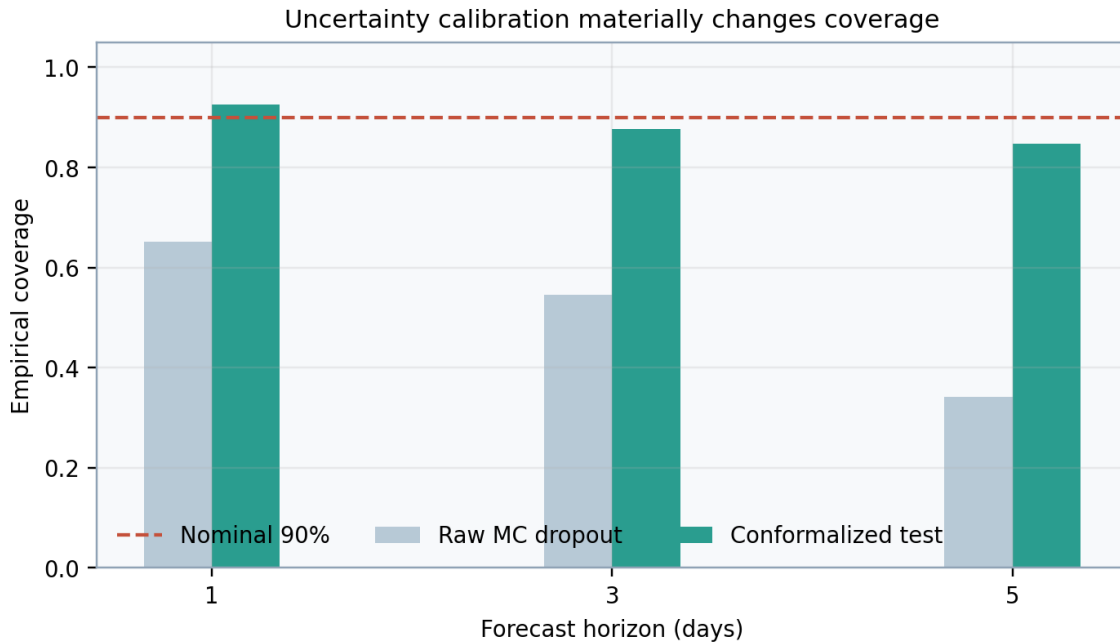


Figure 5. Raw MC-dropout and conformalized test coverage relative to the 90% target.

The conformal procedure improves coverage but does not make every interval a strict distribution-free claim. Exchangeability is challenged by time dependence and nonstationarity, and the deployed radius uses the full validation sample. The repository consequently labels the bounds canonical display/risk bands and sets `intervals_publishable` to false.

## Calibration Findings

| Diagnostic                 | H=1     | H=3     | H=5     | Gate               |
|----------------------------|---------|---------|---------|--------------------|
| Raw MC coverage            | 65.2%   | 54.6%   | 34.1%   | Fails nominal      |
| Half-split validation      | 66.5%   | 68.7%   | 74.3%   | Fails 85% minimum  |
| Full-val radius diagnostic | 83.9%   | 85.2%   | 86.5%   | Descriptive        |
| Independent test coverage  | 92.6%   | 87.6%   | 84.8%   | Passes 80% minimum |
| Mean test width            | 13.42°F | 15.52°F | 14.99°F | Risk-band width    |

The test gate passes because all horizons exceed the configured 80% minimum, although H=5 remains below nominal 90%. The half-split diagnostic fails at all horizons, revealing sensitivity to which validation observations estimate the radius. These are not contradictory results: they answer different questions with different residual samples.

The statistically defensible interpretation is conditional. The bands are useful operational risk indicators at the audited snapshot; they are not guaranteed 90% intervals under arbitrary future distribution shift. A rolling calibration dashboard should track empirical coverage, mean width, interval score, and regime-conditional coverage. Recalibration should be triggered by sustained rather than single-day violations, because realized labels arrive asynchronously across horizons.

**Recommended label:** Use “calibrated risk band (artifact-snapshot test coverage shown)” in the website and forecast logs. Reserve “90% predictive interval” for a prospectively validated protocol with frozen calibration and no post-hoc recentering.

## Forecast Governance

Part 3 maps diagnostic checks to NORMAL, CAUTION, or HOLD. The August 24 run passed nine of nine checks and published NORMAL. Checks cover data freshness, model freshness, schema integrity, forecast-source completeness, physical bounds, cross-horizon spread, persistence deviation, NWS deviation, and BNN calibration state. Critical failures halt publication; warnings retain output with qualification.

| Check family    | Example threshold                          | Current result                |
|-----------------|--------------------------------------------|-------------------------------|
| Freshness       | Data $\leq 2$ days; model $\leq 30$ days   | Pass                          |
| Physical bounds | 32°F to 125°F                              | Pass                          |
| Spread          | $\leq 35^\circ\text{F}$ across horizons    | Pass (2.6°F)                  |
| Persistence     | $\leq 15^\circ\text{F}$ deviation          | Pass; anchored H=3/5 exempted |
| NWS sanity      | soft 3°F; strong 8°F; max 18°F             | Pass                          |
| Calibration     | test coverage $\geq 75\%$ governance floor | Pass with half-split warning  |

Governance improves operational safety by making publication conditional and auditable. Yet rule interactions require scrutiny. Exempting anchored horizons from persistence sanity is logically circular: the post-processor can move a forecast because of NWS and then avoid a check because it was moved. A more robust design should evaluate pre-anchor and post-anchor values independently and require that every policy transformation demonstrate out-of-sample incremental skill.

Publish = NORMAL only if all critical checks pass and warning burden  $<$  policy limit (14)



## Realized Attribution and the Central Audit Finding

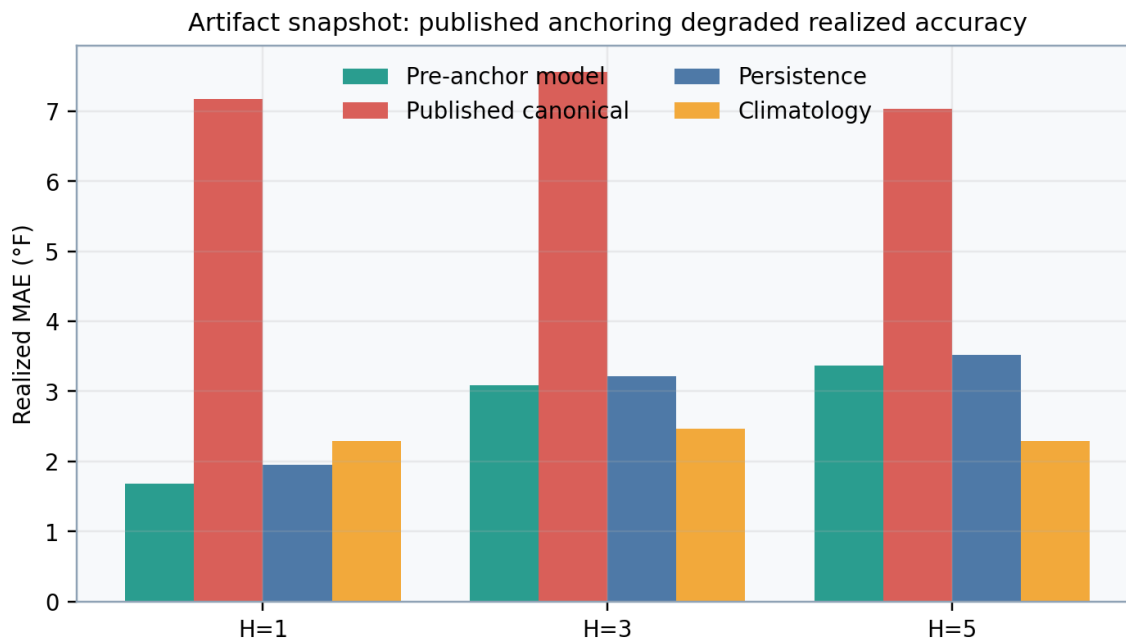


Figure 6. Realized MAE for pre-anchor, published, persistence, and climatology forecasts.

The 65-row attribution tape contains 64, 62, and 60 matured outcomes at H=1, H=3, and H=5.

The pre-anchor model is competitive: 1.68°F, 3.09°F, and 3.37°F MAE. Published forecasts are dramatically worse: 7.18°F, 7.56°F, and 7.04°F. Persistence achieves 1.95°F, 3.22°F, and 3.52°F; climatology achieves 2.29°F, 2.47°F, and 2.29°F.

Thus, the operational layer—not the learned forecast—is the dominant source of error in this historical tape. The finding is strengthened by paired logging because pre- and post-anchor values share the same feature dates and outcomes. It is weakened as a forward-looking claim because the NWS grid point was corrected after much of the tape was generated. Historical NWS comparisons are therefore a diagnosis of the old configuration, not an estimate of the corrected configuration's future skill.

## Spatial Benchmark Failure and Source Contamination

The repository audit identified a hardcoded NWS grid point that was several cells inland from KLAX. The resulting benchmark series showed approximately -8°F bias against airport-centered realized highs: -8.28°F, -8.31°F, and -7.68°F by horizon. NWS anchoring was used on 60 of 65 rows, including 49, 53, and 52 horizon-specific applications. Consequently, the canonical forecast and the NWS baseline are not independent on anchored rows.

$$\text{Bias}(F_h) = (1-a_h)\text{Bias}(B_h) + a_h \text{Bias}(\text{NWS}_h) \quad (15)$$

The algebra explains the artifact. When  $a_h$  is large and the benchmark bias is large, the canonical bias must move toward the benchmark even if  $B_h$  is approximately unbiased. The attribution report explicitly warns that model-versus-NWS skill comparisons are invalid for anchored rows and provides `model_only_metrics` as the unbiased comparison track.

**Current corrective state:** The data artifact now verifies LOX/148,41 by a live points lookup and reports `match=true`. This validates the present spatial mapping. It does not retroactively repair forecasts issued with the old mapping.

The appropriate next experiment is a prospective A/B shadow evaluation: publish or log the unanchored model, corrected-grid anchor, and persistence/climatology baselines for at least one warm season. Freeze anchor rules beforehand and compare paired losses with uncertainty intervals. Until then, the safest operational policy is zero NWS anchor weight with NWS retained as an independent benchmark.

## Statistical Validity, Dependence, and Identifiability

Weather observations are serially dependent, seasonally nonstationary, and exposed to changing upstream model systems. Standard iid interpretations of cross-validation, conformal coverage, and p-values are therefore inappropriate. Chronological holdouts address ordering but still represent one contiguous historical realization. Effective sample size can be far smaller than the nominal number of days, especially within persistent regimes.

$$ESS \approx n / (1 + 2 \sum_{k \geq 1} \rho_k) \quad (16)$$

Model-component identifiability is also limited. LSTM, XGBoost, atmospheric-alpha features, and NWS forecasts share overlapping information about recent temperature and synoptic state. A high blend weight does not identify a causal contribution. Likewise, an HMM state number is permutation-invariant and receives physical meaning only through post-fit diagnostics.

Recommended evaluation uses rolling-origin backtests, block bootstrap uncertainty, proper scoring rules for probabilistic forecasts, and paired loss differences. A Diebold-Mariano-style comparison should use heteroskedasticity/autocorrelation-consistent variance and should be interpreted cautiously for overlapping multi-day targets. For intervals, weighted or adaptive conformal methods are worth studying, but no method removes the need for prospective monitoring under distribution shift.

$$\text{Skill}(\text{reference}) = 1 - \text{MAE}_{\text{model}} / \text{MAE}_{\text{reference}} \quad (17)$$

The audit's negative live skill scores are operationally meaningful because they compare realized dates; however, they should be stratified before generalization by season, regime, heat threshold, and data-source era.

## Software Architecture and Reproducibility

The repository uses modular Python scripts, versioned JSON/parquet/CSV artifacts, GitHub Actions for CI and Pages, and a static interactive index. Validators enforce target clocks, live-tail preservation, split counts, model/schema compatibility, interval display contracts, NWS row-level storage, regime-probability redundancy, and alpha-feature metadata. This is a strong research-engineering foundation because statistical claims are coupled to machine-checkable contracts.

| Layer       | Primary artifact                          | Reproducibility function             |
|-------------|-------------------------------------------|--------------------------------------|
| Data        | historical_daily.parquet; data_meta.json  | Source range, continuity, grid check |
| Features    | feature_matrix.parquet; feature_meta.json | Schema, targets, splits, live tail   |
| Models      | *.pt, *.pkl, part2_meta.json              | Parameters, metrics, hyperparameters |
| Uncertainty | calibration_report.json                   | Coverage, widths, display semantics  |
| Governance  | governance_report.json                    | Checks, thresholds, publish mode     |
| Attribution | live_attribution_report.json              | Matured outcomes and baselines       |

Two improvements would materially strengthen reproducibility. First, every production run should write an immutable manifest containing commit SHA, dependency lock hash, source request timestamps, artifact hashes, and random seeds. Second, generated forecasts should be append-only, with corrections recorded as new versions rather than overwrites. These changes would support forensic reconstruction of any public forecast.

## Recommendations and Conclusion

The laboratory has a credible statistical core but an unsafe historical publication transform. The following recommendations are ordered by expected scientific value. (1) Set NWS anchor weights to zero until a corrected-grid prospective shadow test establishes positive paired skill. (2) Treat NWS as an independent benchmark and preserve row-level issuance values. (3) Promote the XGBoost sleeve only through frozen rolling-origin comparisons, not one split. (4) Report pre-anchor and published forecasts side by side. (5) Add regime-, season-, and heat-event calibration tables with minimum sample gates. (6) Replace display-band language with explicit calibration semantics. (7) adopt immutable run manifests and artifact hashes.

The main substantive conclusion is not that deep learning failed. The committed holdout and live pre-anchor errors show useful predictive structure, especially at  $H=1$ . Instead, a spatially mismatched benchmark and aggressive anchor rules masked that structure. This is a general lesson for operational statistics: post-processing can dominate the error budget, and trusted external forecasts can become a source of correlated bias when their spatial and temporal contracts are not verified.

**Decision:** Retain the modular pipeline, corrected NWS point check, temporal contracts, uncertainty diagnostics, and governance logging. Disable benchmark-driven forecast movement pending prospective evidence. Publish the report and laboratory as an auditable research interface, not as an automated public-safety forecast authority.

With those changes, the project can evolve from a technically rich demonstration into a defensible forecasting experiment whose scientific claims, operational decisions, and web presentation share one traceable evidence chain.

## References

- Angelopoulos, A. N., & Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4), 494–591. <https://doi.org/10.1561/2200000101>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). <https://doi.org/10.1145/2939672.2939785>
- Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. *Proceedings of Machine Learning Research*, 48, 1050–1059. <https://proceedings.mlr.press/v48/gal16.html>
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. <https://doi.org/10.1198/016214506000001437>
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679–688. <https://doi.org/10.1016/j.ijforecast.2006.03.001>
- Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in Neural Information Processing Systems*, 30.

## References (Continued)

- Morelli, S. (2026). Los Angeles Daily Temperature High [Computer software]. GitHub.  
<https://github.com/salemmorelli1/Los-Angeles-Daily-Temperature-High>
- National Weather Service. (2026). API web service. <https://www.weather.gov/documentation/services-web-api>
- Open-Meteo. (2026a). Historical Weather API. <https://open-meteo.com/en/docs/historical-weather-api>
- Open-Meteo. (2026b). Weather Forecast API. <https://open-meteo.com/en/docs>
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., ... Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32.
- Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2), 257–286. <https://doi.org/10.1109/5.18626>
- Rasp, S., & Lerch, S. (2018). Neural networks for postprocessing ensemble weather forecasts. *Monthly Weather Review*, 146(11), 3885–3900. <https://doi.org/10.1175/MWR-D-18-0187.1>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- Wilks, D. S. (2019). *Statistical methods in the atmospheric sciences* (4th ed.). Elsevier.

## Appendix A: Reproduction Code for Figures and Tables

The following Python reads the repository artifacts and reconstructs the principal result tables

without fabricating missing values. Paths are relative to the repository root.

```
from pathlib import Path
import json
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

ROOT = Path(__file__).resolve().parents[1]
meta = json.loads((ROOT/"artifacts_part2/part2_meta.json").read_text())
xgb = json.loads((ROOT/"artifacts_part2b/part2b_summary.json").read_text())
cal = json.loads((ROOT/"artifacts_part2c/calibration_report.json").read_text())
attr = json.loads((ROOT/"artifacts_part9/live_attribution_report.json").read_text())

horizons = [1, 3, 5]
model_rows = []
for h in horizons:
    model_rows.append({
        "horizon": h,
        "lstm_test_mae_f": meta["test_metrics"][f"test_h{h}_mae_f"],
        "xgb_test_mae_f": xgb["test_metrics"][f"h{h}_mae_f"],
        "model_only_live_mae_f": attr["model_only_metrics"][f"h{h}"]["mae_f"],
        "canonical_live_mae_f": attr["metrics_by_horizon"][f"h{h}"]["mae_f"],
    })
model_table = pd.DataFrame(model_rows)
print(model_table.to_markdown(index=False))

coverage_rows = []
for h in horizons:
    coverage_rows.append({
        "horizon": h,
        "raw_mc_coverage": cal["raw_mc_dropout_calibration_results"][f"h{h}_coverage_90pct"],
        "test_conformal_coverage": cal["test_coverage_results"][f"h{h}_coverage_90pct"],
        "mean_width_f": cal["test_coverage_results"][f"h{h}_mean_ci_width_f"],
    })
coverage_table = pd.DataFrame(coverage_rows)
print(coverage_table.to_markdown(index=False))
```

**Reproduction rule:** Run against the same commit and retain the artifact timestamps in any derivative figure caption. Do not merge corrected-grid runs with the historical misaligned-grid tape without an era indicator.



## Appendix A: Figure Construction Code (Continued)

```
# Figure: holdout MAE
x = np.arange(len(horizons)); width = 0.34
fig, ax = plt.subplots(figsize=(7.4, 3.8))
ax.bar(x-width/2, model_table.lstm_test_mae_f, width, label="LSTM")
ax.bar(x+width/2, model_table.xgb_test_mae_f, width, label="XGBoost")
ax.set_xticks(x, [f'H={h}' for h in horizons])
ax.set_ylabel("Test MAE (°F)"); ax.legend(); fig.tight_layout()
fig.savefig("holdout_mae.png", dpi=220)

# Figure: calibration coverage
fig, ax = plt.subplots(figsize=(7.4, 3.8))
ax.bar(x-width/2, coverage_table.raw_mc_coverage, width, label="Raw MC dropout")
ax.bar(x+width/2, coverage_table.test_conformal_coverage, width,
      label="Conformalized test")
ax.axhline(90, color="firebrick", linestyle="--", label="Nominal 90%")
ax.set_ylim(0, 1.05); ax.set_xticks(x, [f'H={h}' for h in horizons])
ax.set_ylabel("Empirical coverage"); ax.legend(); fig.tight_layout()
fig.savefig("interval_coverage.png", dpi=220)

# Figure: operational attribution
persist = [attr["metrics_by_horizon"][f'h{h}"][f'persistence_mae_f'] for h in horizons]
clim = [attr["metrics_by_horizon"][f'h{h}"][f'clim_mae_f'] for h in horizons]
fig, ax = plt.subplots(figsize=(7.6, 4.0))
for offset, values, label in [(-1.5, model_table.model_only_live_mae_f, "Pre-anchor"),
                             (-.5, model_table.canonical_live_mae_f, "Canonical"),
                             (.5, persist, "Persistence"), (1.5, clim, "Climatology")]:
    ax.bar(x+offset*.20, values, .20, label=label)
ax.set_xticks(x, [f'H={h}' for h in horizons]); ax.set_ylabel("MAE (°F)")
ax.legend(ncol=2); fig.tight_layout(); fig.savefig("live_attribution.png", dpi=220)
```

Note. The document-generation script also formats APA tables from these data frames and adds

alternative text to each embedded figure. The browser laboratory is explanatory; the Python artifacts

remain the authoritative computational record.